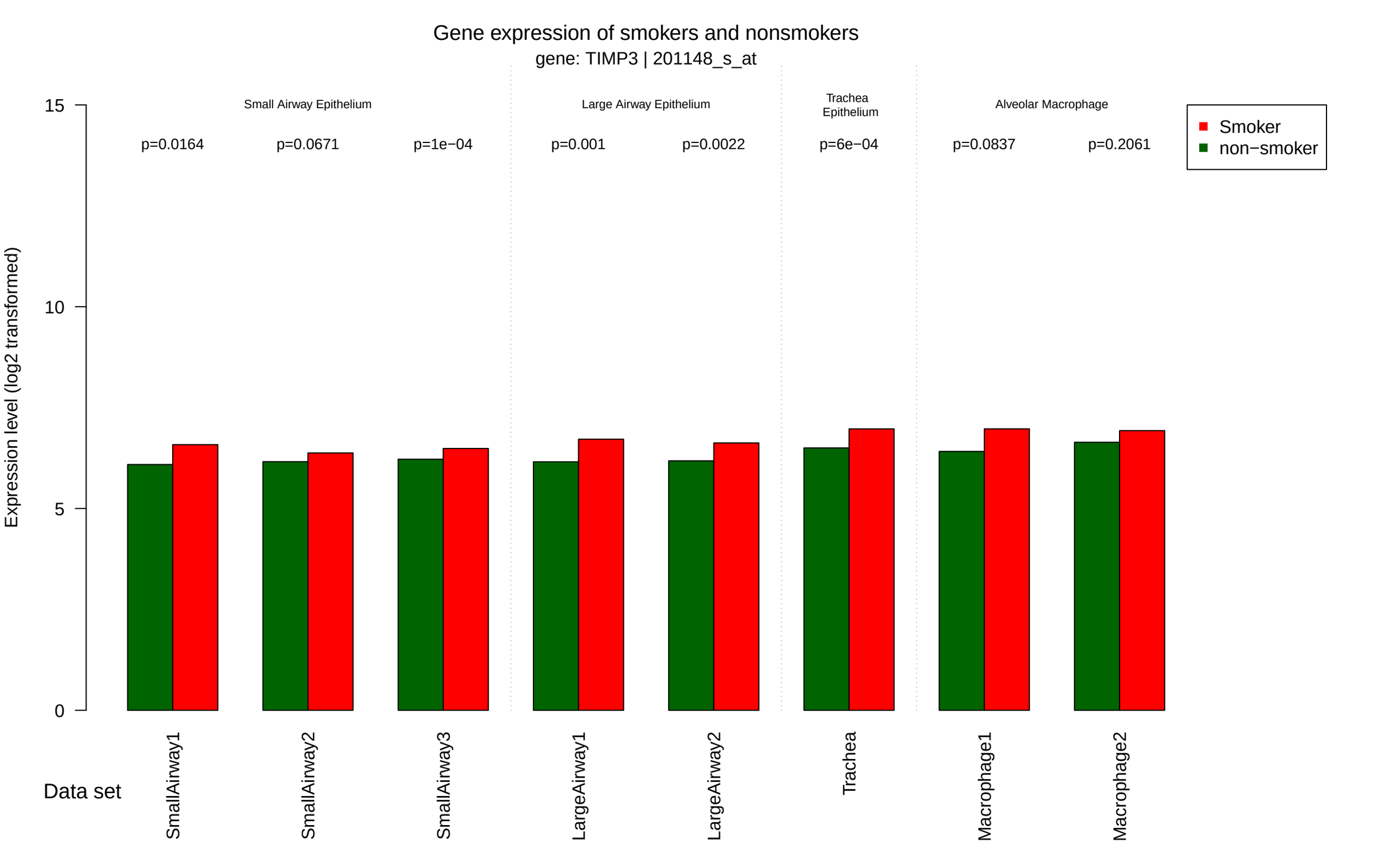
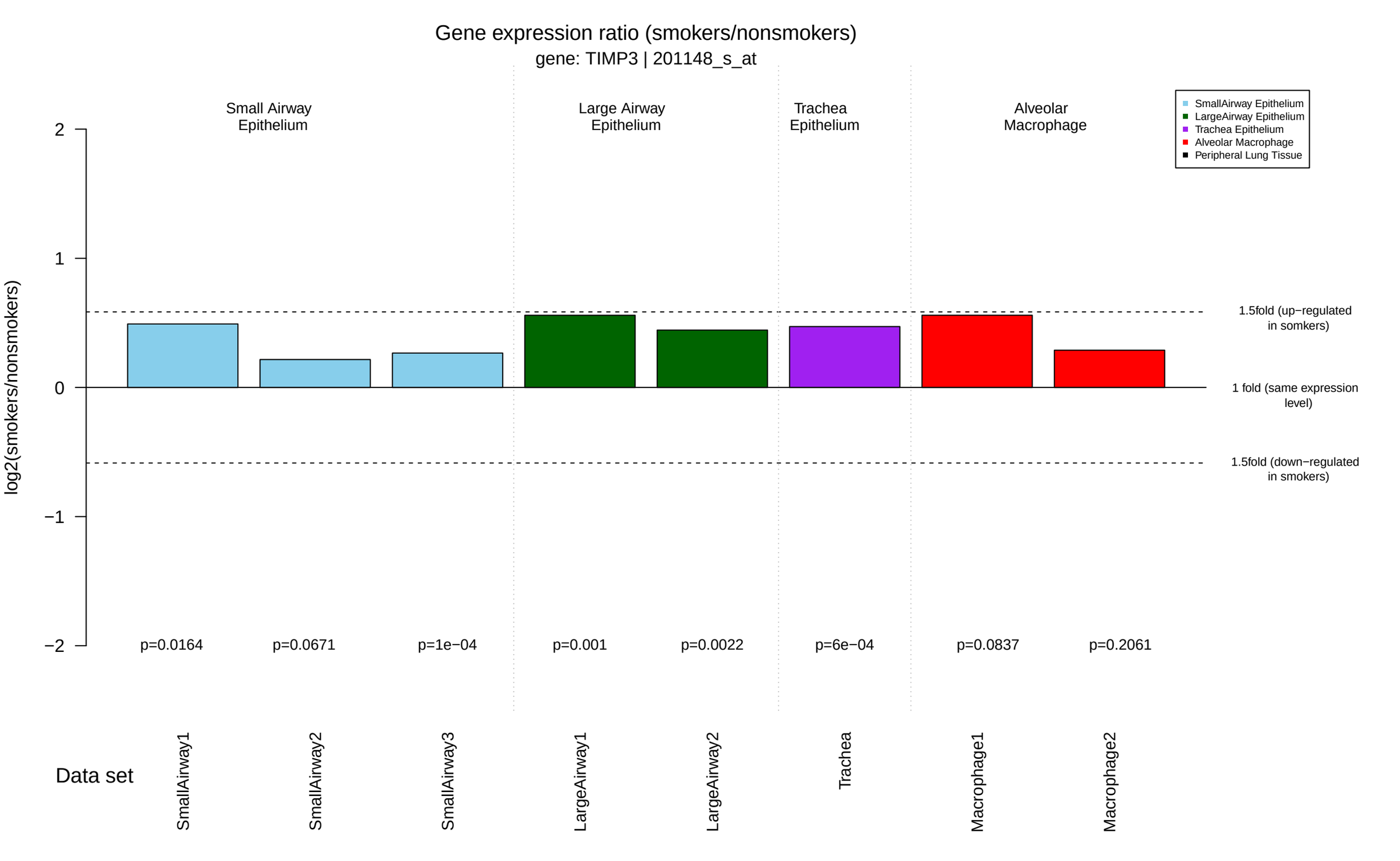
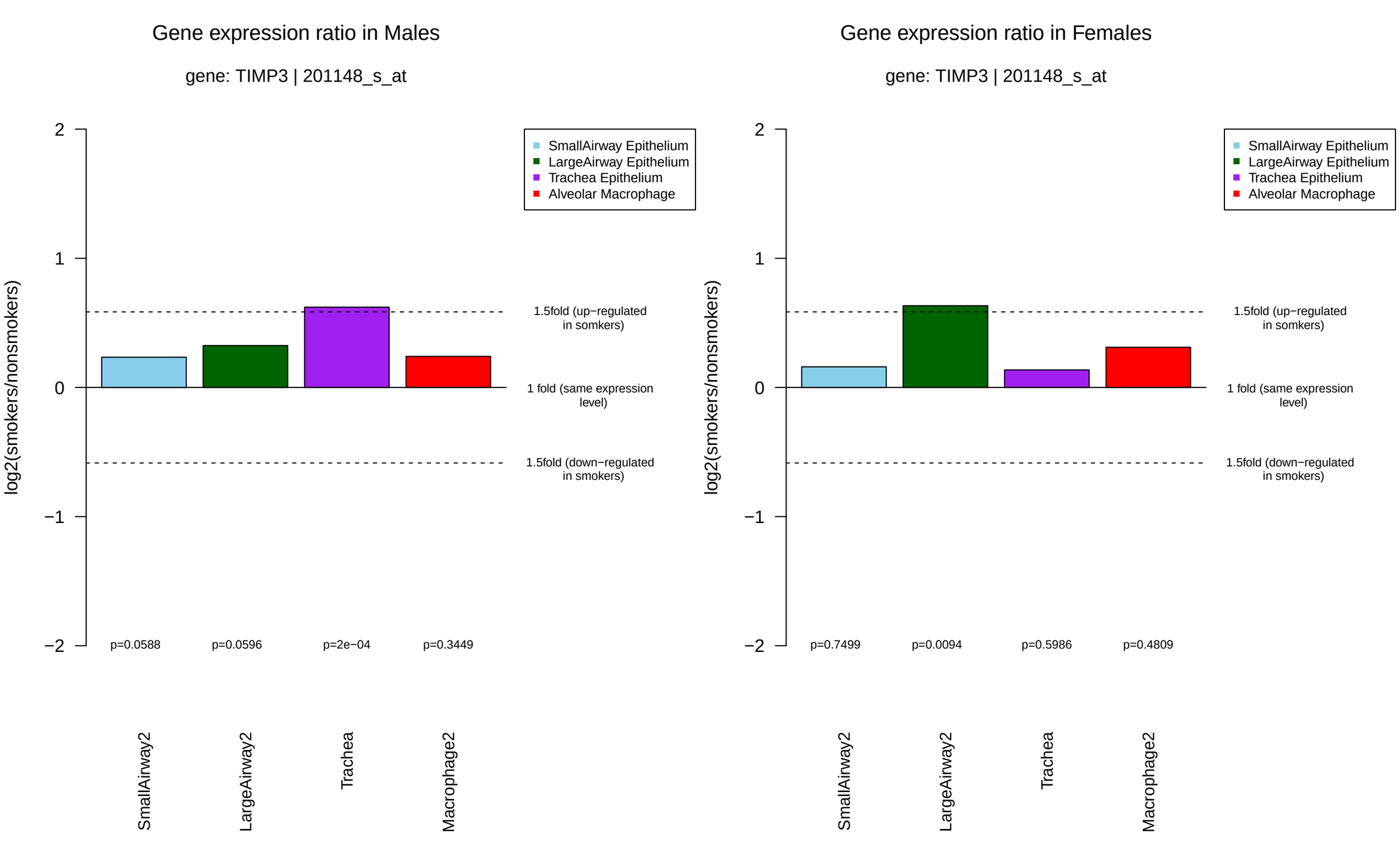
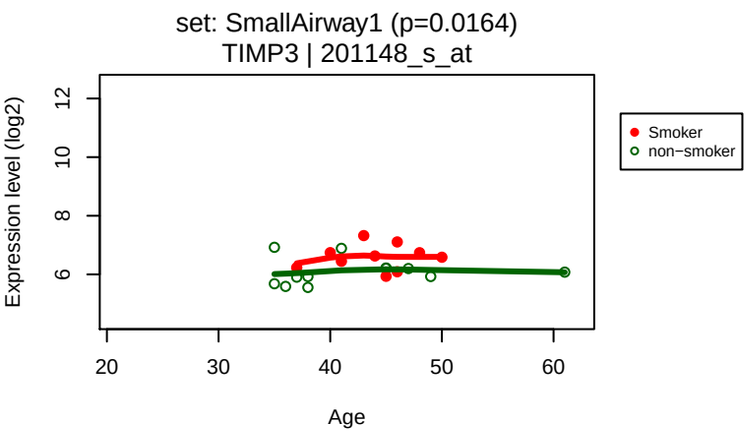
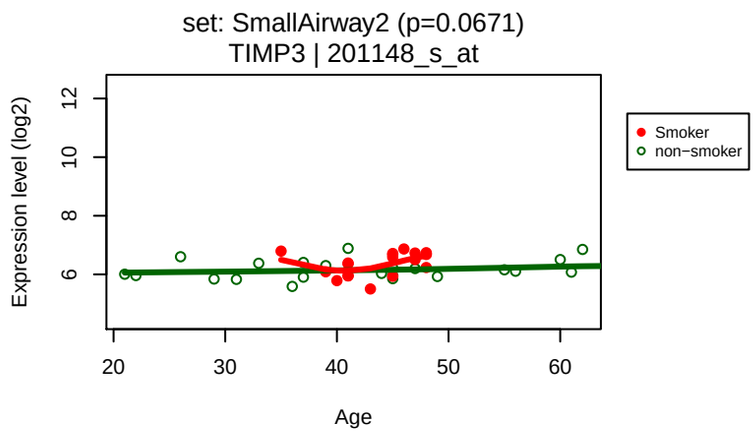
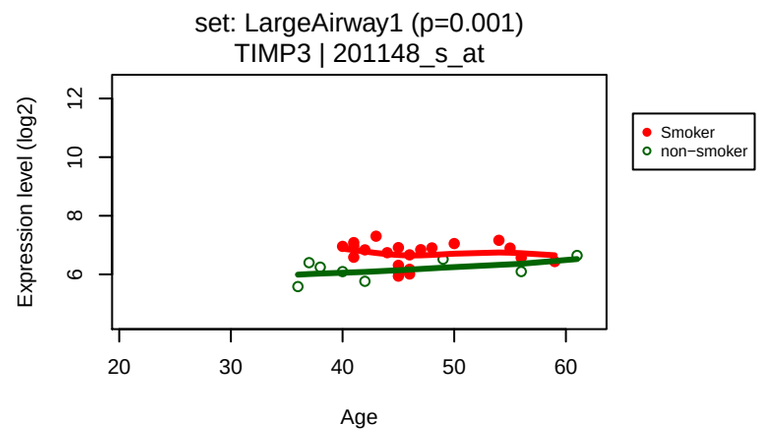

Gene expression of smokers and nonsmokers
gene: TIMP3 | 201148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0164
p=0.0671
p=1e−04
p=0.001
p=0.0022
p=6e−04
p=0.0837
p=0.2061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

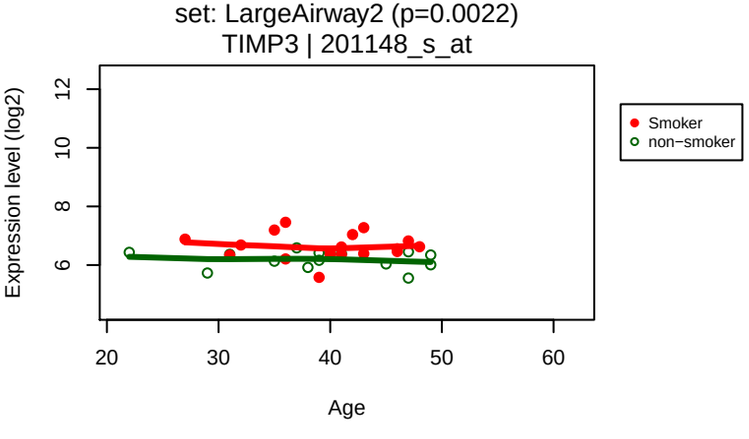

Gene expression ratio (smokers/nonsmokers)
gene: TIMP3 | 201148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0164
p=0.0671
p=1e−04
p=0.001
p=0.0022
p=6e−04
p=0.0837
p=0.2061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

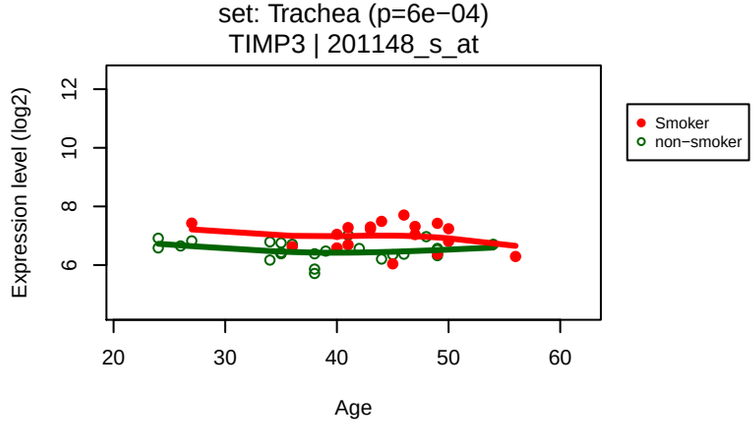

Gene expression ratio in Males
gene: TIMP3 | 201148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0588
p=0.0596
p=2e−04
p=0.3449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP3 | 201148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7499
p=0.0094
p=0.5986
p=0.4809
SmallAirway2
LargeAirway2
Trachea
Macrophage2

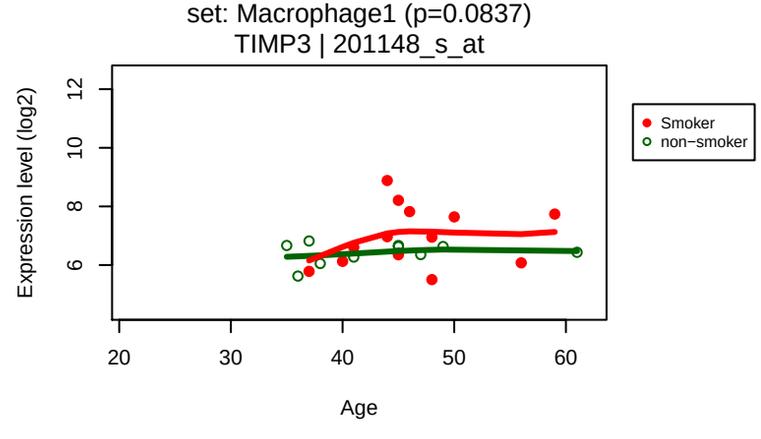

set: SmallAirway1 (p=0.0164)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker

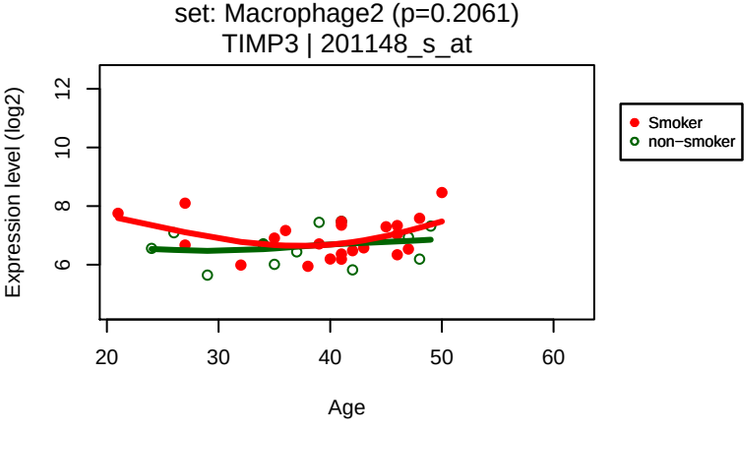

set: SmallAirway2 (p=0.0671)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker

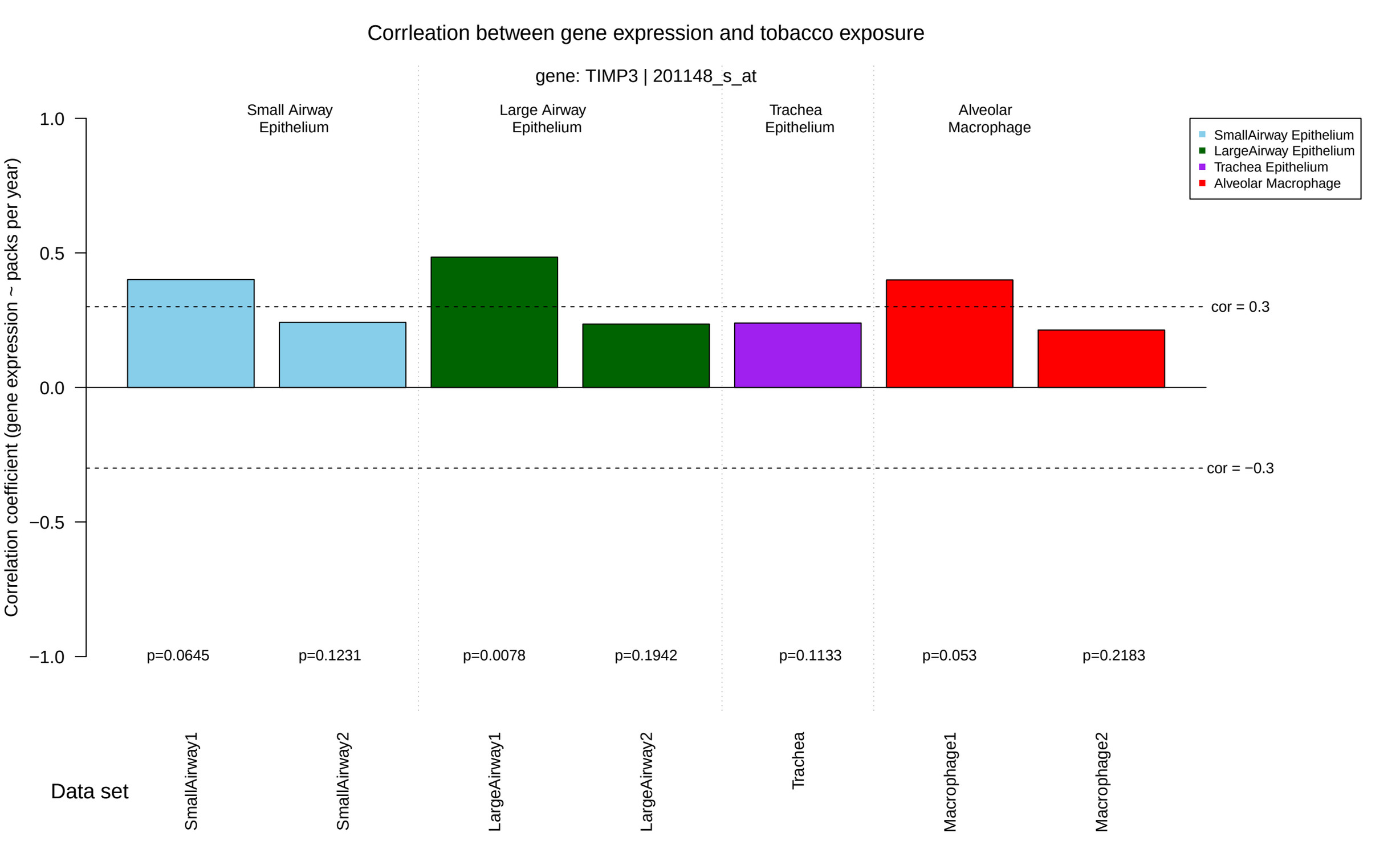

set: LargeAirway1 (p=0.001)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0022)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0837)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2061)
TIMP3 | 201148_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP3 | 201148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0645
p=0.1231
p=0.0078
p=0.1942
p=0.1133
p=0.053
p=0.2183
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2